ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING — SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain 3-Tier Database Architecture and Data Independence. Design ER diagram for Hospital Management. [14 Marks]

[OR]

Q.1 (Alt) Convert ER diagram into Relational Tables. Explain Generalization and Aggregation. [14 Marks]

--- UNIT II ---

Q.2 Explain Relational Algebra fundamental operations. Write complex SQL queries with joins. [14 Marks]

[OR]

Q.2 (Alt) Compare TRC and DRC. Explain Views, Triggers, and Stored Procedures. [14 Marks]

--- UNIT III ---

Q.3 Explain Functional Dependencies and Armstrong Axioms. Decompose into 1NF, 2NF, 3NF, BCNF. [14 Marks]

[OR]

Q.3 (Alt) Explain Lossless Join Decomposition and Dependency Preservation. [14 Marks]

--- UNIT IV ---

Q.4 Explain ACID properties of transactions. Test Conflict Serializability using Precedence Graph. [14 Marks]

[OR]

Q.4 (Alt) Explain Two-Phase Locking (2PL), Strict 2PL, and Timestamp-based ordering. [14 Marks]

--- UNIT V ---

Q.5 Describe Log-Based Recovery and Checkpoints. Compare B-Tree vs B+ Tree indexing. [14 Marks]

[OR]

Q.5 (Alt) Explain Query Optimization pipeline and Deadlock handling in databases. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****